

Edge vs. Cloud: Benchmarking Snapdragon X Elite for On-Device Generative AI Inference

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Abstract

Generative AI workloads are predominantly deployed on cloud GPU infrastructure, providing high throughput but incurring recurring inference costs and network latency. As edge hardware matures, on-device inference presents a compelling alternative for latency-sensitive and cost-constrained applications. In this work, we present a systematic benchmarking study comparing Stable Diffusion 2.1 text-to-image generation across three hardware platforms: the Qualcomm Snapdragon X Elite (edge NPU), NVIDIA Tesla V100 (cloud GPU), and NVIDIA H100 (cloud GPU). We develop an end-to-end benchmarking application with a React/Flask architecture that enables controlled, reproducible comparisons across identical model configurations, prompts, random seeds, and inference parameters. We evaluate each platform along four axes: *latency*, *cost*, *output quality* (measured via CLIP score), and *energy efficiency* (measured in joules per inference). Our results demonstrate that on-device inference on the Snapdragon X Elite achieves perceptual output quality comparable to cloud GPU baselines (CLIP scores within 0.3% of each other), while exhibiting 4.4 $\times$ higher per-image latency relative to the V100 and 16.8 $\times$ relative to the H100. However, the edge device eliminates marginal inference cost entirely, reaching cost parity with the V100 after approximately 531 days and the H100 after only 31 days of continuous operation. Furthermore, the edge NPU consumes 8.6 $\times$ –15.8 $\times$ less energy per inference than cloud GPUs, demonstrating that faster inference does not imply more efficient inference. These findings provide quantitative guidance for practitioners evaluating edge deployment of generative models and inform system-level tradeoff analysis between latency, cost, quality, and energy.

Keywords: edge inference, cloud computing, generative AI, Stable Diffusion, Snapdragon X Elite, benchmarking, quantization, text-to-image generation

1 Introduction

Generative AI has rapidly advanced in recent years, with diffusion-based models achieving state-of-the-art results in text-to-image synthesis [1, 2]. These models are typically deployed on cloud GPU infrastructure, where access to high-bandwidth memory and parallel compute enables fast inference. However, cloud deployment introduces two persistent challenges: recurring per-inference costs that scale linearly with usage, and network latency that degrades user experience in interactive applications.

The emergence of capable edge AI hardware—such

as Qualcomm’s Snapdragon X Elite processor with its integrated Neural Processing Unit (NPU)—raises a natural question: *can on-device inference deliver acceptable quality and latency for generative AI workloads while eliminating cloud infrastructure costs?*

Despite growing interest in edge AI deployment, controlled comparisons between edge and cloud platforms for generative inference remain scarce. Existing benchmarks typically focus on classification or detection tasks, and studies of generative models often compare only cloud-tier GPUs without including edge devices. The performance, cost, and quality characteristics of on-device generative inference are not well characterized

under standardized conditions.

In this work, we address this gap through a systematic benchmarking study. Our contributions are as follows:

1. We develop an end-to-end benchmarking application that enables controlled comparison of Stable Diffusion 2.1 inference across edge and cloud platforms using identical model weights, prompts, seeds, and inference parameters.
2. We evaluate three hardware platforms—Snapdragon X Elite (edge), NVIDIA V100 (cloud), and NVIDIA H100 (cloud)—along four axes: latency, cost, output quality, and energy efficiency.
3. We quantify the cost crossover points at which on-device inference becomes more economical than cloud GPU rental, providing actionable guidance for deployment decisions.
4. We investigate quantization strategies for deploying diffusion models on heterogeneous hardware, including weight quantization via per-output-channel INT8 scaling with dequantization to FP16 for ONNX/TensorRT export.

2 Related Work

2.1 Diffusion Models for Image Generation

Diffusion models [3, 4] have become the dominant paradigm for high-fidelity image generation. Latent Diffusion Models (LDMs) [1], which operate in a compressed latent space, significantly reduce computational requirements while maintaining output quality. Stable Diffusion, an open-source implementation of LDMs, has become a widely adopted benchmark for text-to-image generation and serves as our evaluation model.

2.2 Edge AI Inference

Recent work has explored deploying deep learning models on edge devices for real-time inference [5, 6]. Qualcomm’s AI Engine, leveraging the Hexagon NPU in Snapdragon processors, supports optimized execution of neural network workloads through the Qualcomm Neural Network (QNN) SDK. However, most edge AI benchmarks focus on classification and object detection rather than generative models, leaving a gap in understanding edge capabilities for computationally intensive diffusion-based generation.

2.3 Model Quantization

Quantization reduces model precision from floating-point to lower-bit representations, enabling faster inference and reduced memory footprint on resource-constrained hardware [7, 8]. Post-training quantization

(PTQ) methods, including per-channel weight quantization, have been successfully applied to diffusion models with minimal quality degradation [9, 10]. Our work employs per-output-channel INT8 weight quantization with FP16 activations (W8A16) for the UNet backbone, following best practices for edge deployment.

2.4 Cloud vs. Edge Benchmarking

While extensive benchmarking suites exist for cloud GPU inference [12], comprehensive studies that include edge devices in controlled generative AI evaluations are limited. Our work fills this gap by providing an apples-to-apples comparison framework spanning both deployment paradigms.

3 System Design

3.1 Architecture Overview

We design and implement a full-stack benchmarking application consisting of three components: a React-based frontend dashboard, a Flask backend server, and modular inference backends for edge and cloud execution.

Frontend. The React dashboard provides a unified interface for configuring inference parameters (prompt, negative prompt, number of seeds, guidance scale range, number of diffusion steps), selecting the target device (edge or cloud), and visualizing generated images alongside performance metrics. A side-by-side comparison view displays matched edge and cloud outputs with per-image latency and CLIP scores.

Backend. The Flask server orchestrates inference requests, routes them to the appropriate backend, and aggregates timing and quality metrics. For cloud inference, it communicates with GPU endpoints hosted on the Verda Cloud Platform. For edge inference, it invokes local model execution through the Qualcomm AI Hub SDK with QNN NPU acceleration.

Inference Backends. The cloud backend supports NVIDIA V100 and H100 GPUs with TensorRT-optimized inference engines. The edge backend targets the Snapdragon X Elite’s integrated NPU via Qualcomm’s QNN runtime. Both backends execute the same Stable Diffusion 2.1 model with matched configurations to ensure fair comparison.

3.2 Benchmarking Methodology

To ensure controlled comparisons, we enforce the following invariants across all experiments:

- **Model:** Stable Diffusion v2.1 with identical pretrained weights.
- **Prompts:** A fixed set of 10 diverse prompts spanning multiple visual categories (see Table 1).

- **Seeds:** Identical random seeds per prompt to enable pixel-level comparison.
- **Parameters:** Fixed guidance scale, number of inference steps (20), and output resolution (512×512).
- **Metrics:** Per-image generation latency, CLIP score [11] for text-image alignment, amortized cost per image, and energy consumption per inference (joules).

4 Technical Approach

4.1 Quantization Pipeline

Deploying Stable Diffusion on heterogeneous hardware requires careful quantization to balance inference speed and output quality. Our quantization pipeline targets the UNet component, which dominates inference latency in the diffusion process.

Step 1: Load UNet in FP16. We load the pretrained Stable Diffusion 2.1 pipeline in FP16 precision using the HuggingFace `diffusers` library:

```
StableDiffusionPipeline.from_pretrained(
    ..., torch_dtype=torch.float16)
```

Step 2: Per-Output-Channel INT8 Weight Quantization. We replace all `nn.Linear` layers in the UNet with custom `QuantizedLinear` modules. For each output channel o , we compute a per-channel scale factor:

$$s_o = \frac{\max(|W_{o,:}|)}{127} \quad (1)$$

where $W_{o,:}$ denotes the weights for output channel o . Weights are stored as INT8 with FP16 scale factors. During the forward pass, weights are dequantized on-the-fly to FP16 before computing the linear transformation.

Step 3: ONNX Export. Since ONNX does not natively serialize custom `QuantizedLinear` layers, we convert back to standard `nn.Linear` modules with dequantized FP16 weights before exporting to ONNX (opset 18). The resulting ONNX graph contains standard operators with FP16 weights and activations.

Step 4: TensorRT Engine Build. We parse the ONNX model into a TensorRT network and build an optimized engine with FP16 mode enabled and dynamic batch optimization profiles. We note that TensorRT does not currently support INT16 inference, which motivates FP16 as the deployment precision for cloud GPU execution.

Verification. We verify numerical consistency by running a forward pass through the quantized PyTorch model before export, confirming that FP16 activations with on-the-fly dequantized weights produce outputs consistent with the original FP16 model.

4.2 Edge Execution

On the Snapdragon X Elite, inference is performed through Qualcomm AI Hub, which compiles the model for execution on the Hexagon NPU via the QNN runtime. The model uses W8A16 quantization natively supported by the NPU’s fixed-function accelerators, enabling efficient matrix operations at reduced precision.

4.3 Quality Evaluation

We adopt the CLIP score [11] as our primary quality metric. The CLIP score measures the cosine similarity between the CLIP text embedding of the prompt and the CLIP image embedding of the generated output, providing a proxy for text-image semantic alignment. While not a complete measure of perceptual quality, it enables automated, reproducible quality assessment across thousands of generated images without human annotation.

5 Experimental Setup

5.1 Hardware Platforms

We evaluate three hardware configurations:

- **Snapdragon X Elite (Edge):** Qualcomm’s flagship laptop processor featuring an integrated Hexagon NPU with dedicated AI acceleration. Runs the model locally with W8A16 quantization via QNN.
- **NVIDIA Tesla V100 16GB (Cloud):** A datacenter GPU available on the Verda Cloud Platform at \$0.1381/hour. Runs the model with TensorRT FP16 optimization.
- **NVIDIA H100 SXM5 80GB (Cloud):** NVIDIA’s flagship datacenter GPU available at \$2.290/hour. Runs the model with TensorRT FP16 optimization.

5.2 Prompt Dataset

We curate a set of 10 diverse text prompts spanning multiple visual categories and complexity levels (Table 1). Each prompt is evaluated with 20 diffusion steps and a fixed guidance scale.

Table 1: Evaluation prompt dataset. Prompts span diverse visual categories including scenes, objects, characters, and abstract compositions.

Prompt
Neon cyberpunk street market at night
Watercolor hills with single tree at sunrise
Photorealistic portrait of elderly woman close up
Futuristic robot dog running in meadow
Floating fantasy castle above clouds
Bowl of ramen on wooden table
Astronaut planting flag on Mars
Glass sphere hovering over marble floor
Medieval knight reading in candle lit library
Minimalist geometric hummingbird

6 Results

6.1 Latency Analysis

Table 2 reports per-image generation latency for each prompt across all three hardware platforms at 20 diffusion steps.

Table 2: Per-image generation latency (seconds) at 20 diffusion steps. The Snapdragon X Elite shows consistent latency around 4.14s, while cloud GPUs achieve sub-second and sub-250ms generation.

Prompt (abbrev.)	SD X Elite	V100+TRT	H100+TRT
Neon cyberpunk	4.149	1.530	0.535
Watercolor hills	4.136	0.869	0.213
Portrait elderly	4.141	0.871	0.213
Robot dog	4.148	0.874	0.212
Floating castle	4.142	0.871	0.213
Bowl of ramen	4.139	0.875	0.213
Astronaut Mars	4.146	0.878	0.213
Glass sphere	4.140	0.873	0.220
Medieval knight	4.143	0.877	0.221
Geometric bird	4.144	0.882	0.220
Mean	4.143	0.940	0.247

The Snapdragon X Elite exhibits remarkably consistent latency across all prompts (std. dev. < 0.005s), suggesting that the NPU execution path has minimal prompt-dependent variance. The H100 achieves the lowest latency at 0.247s mean, representing a $16.8\times$ speedup over the edge device. The V100 achieves a $4.4\times$ speedup.

Table 3 reports aggregate performance metrics averaged over 50 generations per device.

Table 3: Aggregate performance comparison averaged over 50 generations. Generation time, save time, CLIP score, and CLIP computation time are reported.

Device	Gen. (s)	Save (s)	CLIP	CLIP (s)
V100	0.966	0.114	0.333	0.022
H100	0.224	0.086	0.334	0.012
SD X Elite	4.291	0.111	0.334	0.122

6.2 Quality Analysis

A central finding of our study is that **output quality is virtually identical across all three platforms**. The CLIP scores for the V100, H100, and Snapdragon X Elite are 0.3331, 0.3342, and 0.3337, respectively—a variance of less than 0.3%. This confirms that the quantized edge model and the FP16 cloud models produce semantically equivalent outputs for the same prompt and seed combination.

Visual inspection of matched edge-cloud image pairs further corroborates this finding. Side-by-side comparisons across all 10 evaluation prompts—including visually complex scenes such as “neon cyberpunk street market at night” and detail-sensitive subjects such as “bowl of ramen on wooden table”—show that edge-generated images are perceptually indistinguishable from cloud-generated images, with only minor pixel-level differences attributable to numerical precision variations in the inference pipeline.

6.3 Cost Analysis

We model cost under two regimes: (1) cloud GPU rental at published hourly rates, and (2) edge hardware as a one-time capital expenditure with zero marginal inference cost.

Cloud costs:

- **V100:** \$0.1381/hour $\rightarrow$ \$3.31/day
- **H100:** \$2.290/hour $\rightarrow$ \$54.96/day

Edge cost: We estimate the Snapdragon X Elite laptop at approximately \$1,760 (one-time purchase), with negligible marginal cost per inference (electricity only).

Crossover analysis. At continuous utilization, the cumulative H100 rental cost exceeds the laptop purchase price after approximately 31 days ($31 \times \$54.96 \approx \$1,704$). The V100 crossover occurs at approximately 531 days ($531 \times \$3.31 \approx \$1,758$). These crossover points shift in favor of edge deployment as utilization increases and in favor of cloud deployment for sporadic or bursty workloads.

Per-image cost. At the measured generation rates, the V100 produces approximately 3,730 images/hour (at 0.966s/image), yielding a per-image cost of \$0.0000370. The H100 produces approximately 16,070 images/hour,

at \$0.000143/image. Notably, despite being $4.3\times$ faster per image, the H100 incurs a $3.9\times$ higher per-image cost than the V100 due to its substantially higher hourly rental rate. While per-image cloud costs are extremely low in both cases, they accumulate indefinitely, whereas edge costs are bounded by the initial hardware investment.

6.4 Energy Analysis

Beyond latency and monetary cost, energy consumption per inference is a critical dimension for evaluating deployment strategies—particularly at scale, where aggregate energy determines both operational cost and environmental impact. We measure the energy consumed per completed inference on each platform, accounting for the total system power draw during generation.

Our measurements reveal a striking disparity: the Snapdragon X Elite NPU consumes approximately 18 joules per inference, compared to approximately 155 joules for the H100 and approximately 285 joules for the V100. The edge device is thus $8.6\times$ more energy-efficient per inference than the H100 and $15.8\times$ more efficient than the V100, despite exhibiting higher latency.

This result underscores that **faster inference does not imply more efficient inference**. The H100 delivers $\sim 10\times$ lower latency than the edge NPU, but each inference consumes an order of magnitude more energy. The V100, while slower than the H100, is the least energy-efficient of all three platforms due to its older architecture and higher power draw relative to throughput.

An important framing distinction applies here: we are not comparing one edge device against one datacenter GPU in isolation. Rather, we are comparing *distributed inference at the edge*—where each user’s device performs its own generation—against *centralized inference in the datacenter*, where GPUs must remain powered and provisioned to serve requests. Cloud GPUs must stay powered to provide capacity regardless of instantaneous demand, whereas edge NPUs consume energy only when inference actually occurs. When inference is distributed across edge devices, total energy scales linearly with actual usage rather than with provisioned capacity.

This distinction has significant implications for deployment at scale. For workloads characterized by interactive, low-frequency, or privacy-sensitive usage patterns, distributed edge NPUs can significantly reduce aggregate inference energy by eliminating the baseline datacenter overhead associated with maintaining always-on GPU infrastructure. This energy advantage complements the cost and privacy benefits identified in the preceding sections, though it does not suggest that edge deployment replaces cloud for all use cases.

7 Discussion

7.1 Tradeoff Summary

Our results reveal a clear four-way tradeoff:

Latency: Cloud GPUs dominate, with the H100 offering $16.8\times$ lower latency than the Snapdragon X Elite. For latency-critical applications requiring sub-second generation (e.g., interactive creative tools), cloud deployment remains necessary.

Cost: Edge deployment eliminates recurring costs after the initial hardware investment. For sustained workloads, the edge becomes more economical after 31–531 days depending on the cloud GPU tier. For intermittent use, cloud pay-per-use remains more efficient.

Quality: Output quality is effectively equivalent across all platforms, with CLIP score variance below 0.3%. This is a strong result for edge deployment, indicating that quantization and NPU execution do not meaningfully degrade generative output quality for Stable Diffusion 2.1.

Energy: The Snapdragon X Elite consumes $8.6\times$ – $15.8\times$ less energy per inference than cloud GPUs, demonstrating that lower latency does not equate to higher efficiency. For distributed, low-frequency workloads, edge inference offers substantial energy savings by scaling consumption with actual usage rather than provisioned capacity.

7.2 Additional Considerations

Beyond the three primary axes, edge inference offers several additional benefits not captured in our quantitative evaluation. These include data privacy (prompts and outputs never leave the device), offline capability (no network dependency), and predictable latency (no variance from network congestion or cloud scheduling).

Conversely, cloud deployment offers scalability (easy horizontal scaling for batch workloads), access to the latest GPU hardware without capital expenditure, and the ability to run larger models that exceed edge device memory constraints.

8 Limitations and Future Work

Our study has several limitations that motivate future investigation.

Quantization precision. TensorRT does not currently support INT16 inference, constraining our cloud baseline to FP16. Future work should explore inference engines that support INT16 to enable more direct comparison with the edge device’s W8A16 execution.

Quality metrics. While CLIP score provides a useful automated metric for text-image alignment, it does

not fully capture perceptual quality, fine-grained detail fidelity, or aesthetic preference. Incorporating additional metrics such as Fréchet Inception Distance (FID) [13] and human evaluation studies would provide a more complete quality assessment.

Model scope. Our evaluation is limited to Stable Diffusion 2.1. Extending the benchmarking framework to larger state-of-the-art models (e.g., SDXL, Stable Diffusion 3) would test the scalability of edge inference to models with higher computational demands.

Task diversity. Beyond text-to-image generation, future work should evaluate other generative modalities on edge hardware, including large language model inference, text-to-video generation, and image-to-image translation.

Thermal and power analysis. While we report per-inference energy consumption, edge device performance may degrade under sustained thermal load. Long-running benchmarks with continuous thermal monitoring would provide a more realistic assessment of whether the measured energy efficiency holds under production-level sustained workloads.

Utilization assumptions. Our cost crossover analysis assumes continuous utilization, which represents an upper bound on edge cost efficiency. Real-world deployment patterns with intermittent usage would extend the crossover period, and future work should model cost under realistic utilization distributions.

9 Conclusion

We present a controlled benchmarking study comparing Stable Diffusion 2.1 inference across the Qualcomm Snapdragon X Elite edge platform and NVIDIA V100/H100 cloud GPUs. Our end-to-end benchmarking application enables reproducible, apples-to-apples comparisons using identical models, prompts, seeds, and parameters.

Our key finding is that on-device inference achieves output quality indistinguishable from cloud baselines (CLIP score within 0.3%), while exhibiting $4.4\times$ – $16.8\times$ higher latency. Crucially, edge deployment eliminates marginal inference cost, achieving total cost parity with cloud rental within 31–531 days depending on the GPU tier. Moreover, the edge NPU consumes an order of magnitude less energy per inference than cloud GPUs, highlighting that faster inference does not equate to more efficient inference and that distributed edge deployment can significantly reduce aggregate energy consumption for workloads that do not require centralized throughput. These results demonstrate that edge generative AI is viable for applications where cost efficiency, energy efficiency, data privacy, and offline operation are prioritized over raw latency performance.

This work provides quantitative evidence to inform practitioners’ deployment decisions and establishes a methodology for systematic edge-vs-cloud evaluation that can be extended to future models and hardware platforms.

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